

Λ hyperon spin-spin correlations in Run12 $p+p$ collisions at $\sqrt{s} = 200$ GeV

Code guide

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1 Introduction

This document contains a description of all relevant codes and software used in STAR Run12 Λ hyperon spin-spin correlations in Run12 $p+p$ collisions at $\sqrt{s} = 200$ GeV analysis. Each section provides information on how to run the code and where each given code is located, and description of individual source files and input/output files. In case of questions, contact me at jvanek@bnl.gov.

To download the analysis codes do:

```
cvs checkout offline/paper/psn0848
```

The codes for data analysis in Sec. 2 are all written to work on STAR RCF. The codes for simulations from Sec. 3 and for calculation of the spin-spin correlation in Sec. 4 were all used locally with ROOT 6. Functionality on RCF is therefore not guaranteed and it is recommended to run those on your local computer with ROOT 6.

2 Data analysis

2.1 Production

This section describes how to run the code to read PicoDst files and store custom treens with information about single Λ and $\bar{\Lambda}$ hyperons that are later used for the spin-spin correlation analysis. The codes are in the repository in folder called "Production".

First, you need to prepare folders needed for submission of the jobs:

```
./prepare_folders.sh
```

After that, compile the code:

```
./compile_SL21d.csh
```

Then you will need to get the latest file list for this dataset. You can do that by going to folder "picoLists" and running a script that will generate the needed file list:

```
cd picoLists  
./getFileList.sh
```

This will generate a file called "picoList_Run12_SL21d.list" in the correct format for the submission of the jobs.

Now, you should be ready to submit the jobs. You now have two options for the test submission. One is on two PicoDst files I have downloaded to my local folder on RCF. These will always be the same and can be used as a direct cross-check. Second option is use the generated file list, which may change over time as file availability also may change. The submission scripts is located in the "Production" folder.

To submit the small test submit, using just the two local files, you can do:

```
./submit_Run12_SL21d_local.sh
```

The needed file list for this submission is part of the CVS repository.

To submit jobs using the generated file list, you can do:

```
./submit_Run12_SL21d.sh
```

which will submit first 1000 files from the generated file list. Submitting the full data set would be too time consuming, but if you want to submit more files, you can modify the aforementioned submission script and change the file range to be submitted:

```
sed -n '1,1000 p' ./picoLists/picoList_Run12_SL21d.list > picoList_submit.list
```

The submission during the analysis was split into 6 batches in order to keep the number of submitted jobs reasonable. One important note here is that the jobs may run for quite a long time (more than 24 hours).

The jobs will generate output files in

```
Production/production/"date-time"
```

where "date-time" is automatically generated folder based on date and time of job submission. If you decide to submit more than once, please wait for a couple of minutes between submissions so that each submission has its own output folder. In order to compare to a reference output file, go to the aforementioned folder and merge the root files into one:

```
hadd output.root *.root
```

The reference files from my submission of the corresponding test jobs are available on RCF here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Production/production
```

Here you will find two sub-folders, one with output from the two local PicoDst files and one from the 1000 PicoDst files. You should be able to re-produce the output from two local files exactly. The file with 1000 PicoDsts can differ due to varying availability of PicoDst files on the distributed disk.

3 Simulations

This section provides description on how to run and analyze PYTHIA simulation that was used for the following:

- Closure test to verify performance of the mixed-event correction.
- Calculate the systematic uncertainty of mixed-event correction.

3.1 PYTHIA 8.3 simulation

The core of the simulation studies is a standalone PYTHIA 8.3 simulation. The key note here is that this code will not work on RCF, as PYTHIA 8.3 is not available there. So if you want to check this code, you will need to find a system with PYTHIA 8.3 installed, or install it yourself on your local computer. As this is system specific, you need to follow instructions on the official PYTHIA web page <https://www.pythia.org>. You will have to make sure to properly link ROOT to the PYTHIA installation. No other external software is required. When you have working PYTHIA 8.3 installed, you can compile and run the simulation. You will need your own "Makefile.inc" available so that you can compile the macro.

The code itself is located in "/Simulation/PYTHIA". First step is to prepare folders for the code:

```
./prepare_filders.sh
```

Then you will need to copy your "Makefile.inc" into "Lambda_MB" and "K0s_MB". Then you can go to folder "Lambda_MB" and compile the macro:

```
make main
```

and then you should be able to run the code:

```
./run_main.root
```

This will run a sample of the simulation with 100 events. Simulating the full dataset would be too time consuming during the code QA, so only small sample is generated here. A reference file for the small sample is available here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Simulation/PYTHIA/Lambda_MB/output/pp_200/output_Lambda_pp_200_MB_test.root
```

The same procedure needs to be repeated for the K_s^0 simulation. For that go to "K0s_MB" folder and repeat the steps described above. The reference output for the small sample is here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Simulation/PYTHIA/K0s_MB/output/pp_200/output_K0s_pp_200_MB_test.root
```

The full sample for both Λ and K_s^0 is also available on RCF and can be used in the next step of the Code QA. Details are described in the next section.

3.2 Reading PYTHIA simulation

This section describes how to run codes, that read the output from the standalone PYTHIA 8.3 simulation. These codes will save needed distributions from the customs trees generated in the previous step. The code for Λ and K_s^0 works the same way. The Λ code is located in the CVS repository in:

```
/Simulation/PYTHIA/Lambda_MB/Read_PYTHIA_tree
```

and similarly the code for K_s^0 is located in:

```
/Simulation/PYTHIA/K0s_MB/Read_PYTHIA_tree
```

Both of these codes are setup to run on my full simulation files. The CVS repository contains file lists that point to my folder with the input files. The sample size is around 70 GB, so it will probably be better, if you don't copy those to your folder. In the current setup, you can simply run the codes and the files will be loaded correctly.

The following instructions are the same for Λ and K_s^0 . You now have two options how to run. You can run a test run on one file from the full simulation production:

```
./run_test.sh
```

This should not take too long to run (maybe a couple of minutes). You can also read the whole simulation sample by running:

```
./run.sh
```

Note here is that running over the full simulation may take some time to run through and might be beyond scope of the Code QA. The test sample should hopefully be enough to check the codes.

The output from this code for Λ will be located here:

```
/Simulation/PYTHIA/Lambda_MB/output/pp_200/new_out_hist/
```

or for K_s^0 here:

```
/Simulation/PYTHIA/K0s_MB/output/pp_200/new_out_hist/
```

There should be two files - one for the reference test, with one input file (name includes "test"), and second file for the full data-set.

Reference files, generated by me are available on RCF here for Λ :

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Simulation/PYTHIA/Lambda_MB/output/pp_200/new_out_hist
```

and here for K_s^0 :

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Simulation/PYTHIA/K0s_MB/output/pp_200/new_out_hist
```

3.3 Closure test and systematic uncertainty of ME

The next step of the analysis of the simulations is to use the output from the previous step to plot the closure test plots and to store the systematic uncertainty of the mixed event (sys. uncert. from the closure test). The codes to do this step are located here:

```
/Simulation/Closure_test/
```

The first step is to prepare all needed folders by running:

```
./prepare_folders.sh
```

Then you need to copy input to the "input" folder. The inputs in question are files generated in the previous section in

```
/Simulation/PYTHIA/Lambda_MB/output/pp_200/new_out_hist/  
/Simulation/PYTHIA/K0s_MB/output/pp_200/new_out_hist/
```

So simply copy the four files from paths above into the "input" folder for the Closure_test.
Then you should be all set to run the code. To run on the test outputs you can do:

```
./run_test.sh
```

and to run on the files with the full simulation statistics, do:

```
./run.sh
```

This will generate outputs in the "output" folder. There will be files for both the test sample and the full sample. The reference output files generated by me are available on RCF here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/Simulation/  
/Closure_test/output
```

All of these outputs will be important for the next steps of the analysis.

At the same time, the codes produces a series of plots. Note that the plots will be overwritten every time you run the code, regardless of the mode (test or full). If you want to save the plots, please copy them to a different location. My plots for both test and full samples are available on RCF here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/Simulation/  
/Closure_test/figs
```

where you will find three sub-folders. Two of them are for the test and full simulation samples ("test_sample" and "full_sample"). The third is the working output folder for my local copy of the code and you can ignore it.

4 Spin-spin correlation analysis

This section describes how to run codes to extract the spin-spin correlation and generate the paper plots. As stated in the introduction, these codes are generally made to work with ROOT 6 and may not work properly on RC with ROOT 5.

Second note here is, that these codes generally have to be run with the full data-set, as the statistics of the Λ pairs is low. The reference files generated in Sec. 2 don't have sufficient statistics to fit the M_{inv} histograms and will therefore not work with the code described below. Please, use my full statistics input file with data to test this code. Details are provided in the following subsection.

4.1 Analysis

This set of codes is used to analyze the custom trees generated by the code described in Sec. 2. It also uses input files from the simulation described in Sec. 3. So, the first step you need to do is setup all needed folders and make sure you have all needed input files. The codes are located in the CVS repository in folder called "Analysis".

Similar to the other codes, first you need to setup all of the input and output folders by running:

```
./prepare_folders.sh
```

Then you need to get the root files with the custom trees and copy them into:

```
/Analysis/input/data/
```

You can find the input files on RCF here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Analysis/input/data
```

There are 6 files you need to copy and they are about 8 GB in total, so make sure you have enough disk space available. Next, inside of the "input/data" folder generate a file list of your copy of the files:

```
ls $PWD/*.root > fileList.list
```

Here, make sure that the file list has this exact name (fileList.list) and that it is inside of the "input/data" folder.

Another input you need to run these codes is the systematic error of the ME correction, that is generated by the "Closure_test" code described in Sec. 3.3. Here, you need to make sure that you have the following two files in the path below:

```
/Simulation/Closure_test/output/SysErr/SysErrSlope.root  
/Simulation/Closure_test/output/SysErr/SysErrSlope_K0s.root
```

These files should exist from you running the Closure test macros. Please, use the files above and not the test files.

With these inputs in place, you should be ready to run the codes. There is a set of scripts to run everything in the right order and with the correct parameters. Please, run the macros in the order described here, as some of the macros generate files that are needed by other macros. It is important to note here, that running over the data-set with the full mixed-event takes a long time - probably about 2h. This is normal behavior, but you will need to run these long scripts with different settings: tight topological cuts, tight p_T cuts, analysis cuts, all three separately for Λ and for K_s^0 . So prepare enough time for this part of the QA with the full sample.

To allow quicker test of the codes, there is a set of scripts that will run the code with reduced mixed-event size. The test sample codes should run for about 10-15 minutes per script, that read the input trees (macros 1, 2, and 3 for Λ , and macros 4, 5, and 6 for K_s^0). To run in the test mode for Λ do the following set of commands from the "Analysis" folder, in this order:

```
./scripts/1_run_Lambda_tight_topo_cuts_test.sh  
./scripts/2_run_Lambda_tight_pT_cuts_test.sh  
./scripts/3_run_Lambda_ana_cuts_test.sh  
./scripts/4_run_Lambda_get_corr.sh
```

To run the K_s^0 codes in test mode do:

```
./scripts/5_run_K0s_tight_topo_cuts_test.sh  
./scripts/6_run_K0s_tight_pT_cuts_test.sh  
./scripts/7_run_K0s_ana_cuts_test.sh  
./scripts/8_run_K0s_get_corr.sh
```

These codes then generate multiple output files and sets of figures. The output files will be stored in the `"/Analysis/output/"` folder, where you will find three sub-folders:

```
/Analysis/output/invariant_mass/  
/Analysis/output/Polarization/  
/Analysis/output/ProdPlane/
```

My reference outputs for the test sample with small mixed event are available here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Analysis/output/out_test
```

The figures generated by this set of codes can be found `"/Analysis/plots/"`. Please, browse the sub-folders with the individual distributions. My reference plots for the test sample are here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Analysis/plots/plots_test
```

All of these figures are only working plots that are used to check that the framework is working. The paper plots are generated by a separate set of macros that will be described later.

You can also run everything with the full mixed-event, but again, keep in mind that the codes that read the trees will each run for a long time. Also, re-running will re-write all of the output files and figures. To run with the full mixed event, you can just do (same as before, but use scripts without "test" in the name):

```
./scripts/1_run_Lambda_tight_topo_cuts.sh  
./scripts/2_run_Lambda_tight_pT_cuts.sh  
./scripts/3_run_Lambda_ana_cuts.sh  
./scripts/4_run_Lambda_get_corr.sh
```

To run the K_s^0 codes in test mode do:

```
./scripts/5_run_K0s_tight_topo_cuts.sh  
./scripts/6_run_K0s_tight_pT_cuts.sh  
./scripts/7_run_K0s_ana_cuts.sh  
./scripts/8_run_K0s_get_corr.sh
```

The reference output files for the full mixed event are located here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Analysis/output/out_full
```

The reference figures for the full mixed event are here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Analysis/plots/plots_full
```

4.2 Paper plots

The final step in running of the codes is to plot all of the paper plots. The codes are located in the CVS repository in the "Paper_plots" folder. Again, the first step is to prepare the folders needed by the macros:

```
./prepare_folders.sh
```

Next you will need to copy all needed input files into the "input" folder. There are total input 9 files which you should be able to copy from the previous step of the Code QA. Alternatively, you can use my reference files. Below are copy commands that should work, provided you have everything in one directory. From the "Paper_plots" folder do:

```
cp ../Analysis/output/invariant_mass/2012/InvMass_Lambda_ana_cuts.root .

cp ../Analysis/output/Polarization/2012/Polarization_Delta_phi.root .
cp ../Analysis/output/Polarization/2012/Polarization_Delta_y_Delta_phi.root .
cp ../Analysis/output/Polarization/2012/Polarization_Delta_y.root.root .

cp ../Analysis/output/Polarization/2012/Polarization_K0s_Delta_phi.root .
cp ../Analysis/output/Polarization/2012/Polarization_K0s_Delta_y_Delta_phi.root .
cp ../Analysis/output/Polarization/2012/Polarization_K0s_Delta_y.root .

cp ../Simulation/Closure_test/output/polarization/Polarization_PYTHIA.root .
cp ../Simulation/Closure_test/output/polarization/Polarization_K0s_PYTHIA.root .
```

When you have all of the input files copied, just run:

```
./run.sh
```

This will generate all of the plots in the "figs" folder.

The reference inputs, generated by me, which were used to plot the final paper plots are available here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Paper_plots/input/full
```

The final plots are then available here:

```
/star/u/vanekjan/pwg/vanekjan/CodeQA/L_spin_spin_correlations/codes/  
/Paper_plots/figs
```